

Nucleic Acids

Lecturer 3

Presented By

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Chapter II

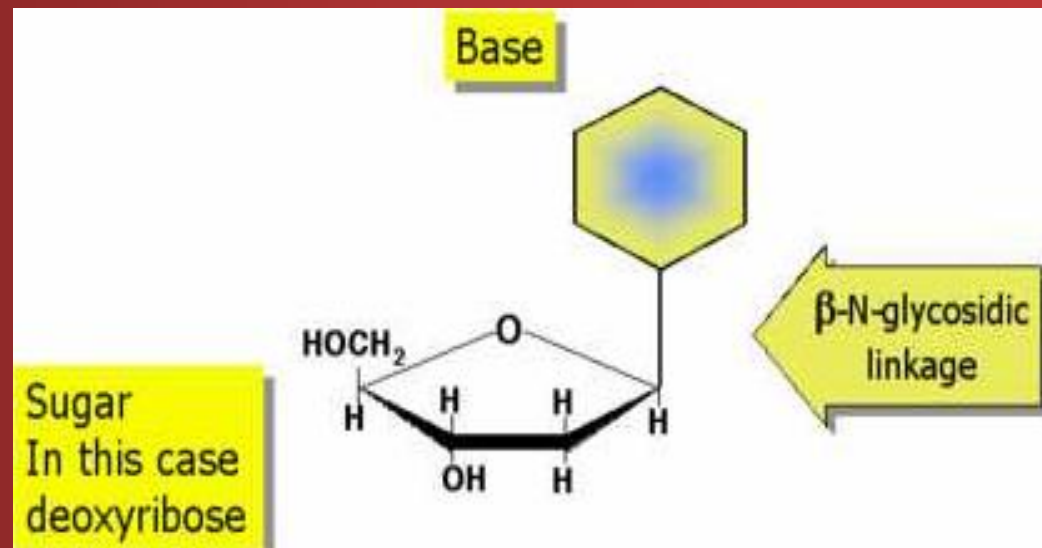
Nucleotides

Chapter Two: Nucleotides

- 1- Nucleotides definition ✓
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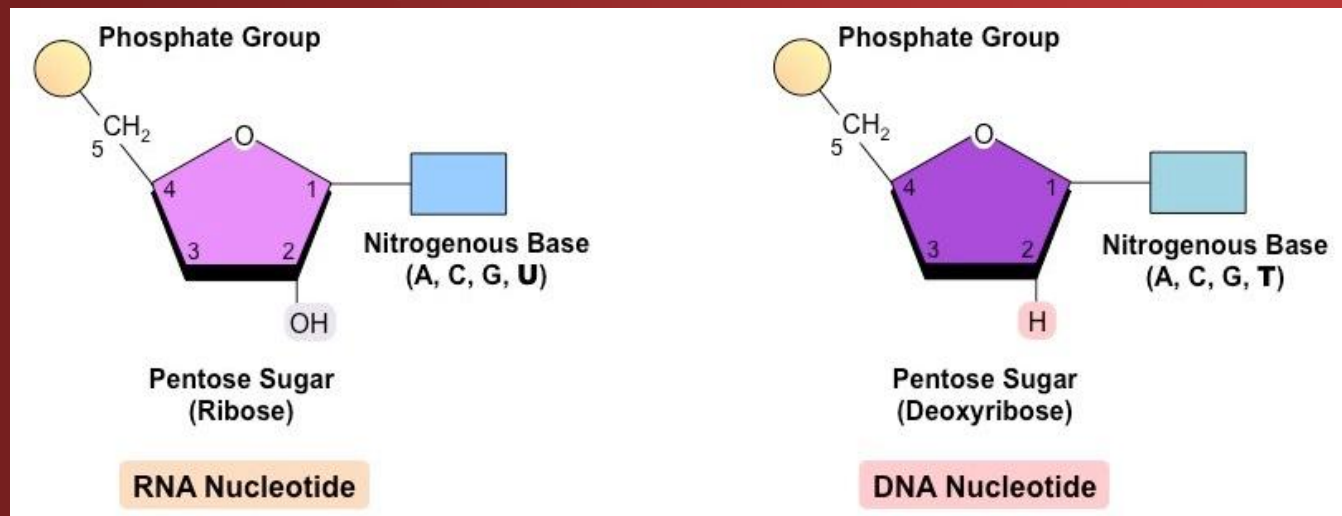
3- Nucleoside structure

- The resulting compound when a purine or a pyrimidine base is linked to ribose or deoxyribose is known as a nucleoside.
- The nucleosides from ribose = ribonucleosides: Adenine forms Adenosine, Guanine forms Guanosine, Cytosine forms Cytidine and Uracil forms Uridine.
- The nucleosides from 2-deoxyribose = deoxyribonucleosides (deoxyadenosine, deoxyguanosine, deoxycytidine, deoxythymidine etc.)



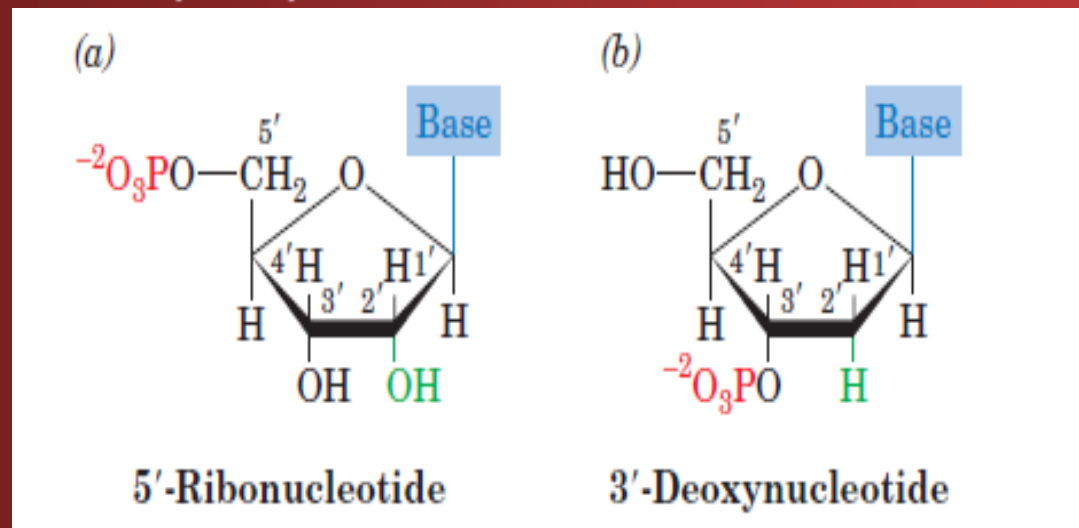
4-Nucleotide structure

- The resulting compound when a purine or a pyrimidine is linked to ribose or deoxyribose and phosphoric acid is known as a nucleotide. Nucleotides are phosphoric acid esters of nucleosides.
- Those derived from ribonucleosides are referred to as ribonucleotides. Those derived from deoxyribonucleosides are referred to as deoxyribonucleotides.



4-Nucleotide structure

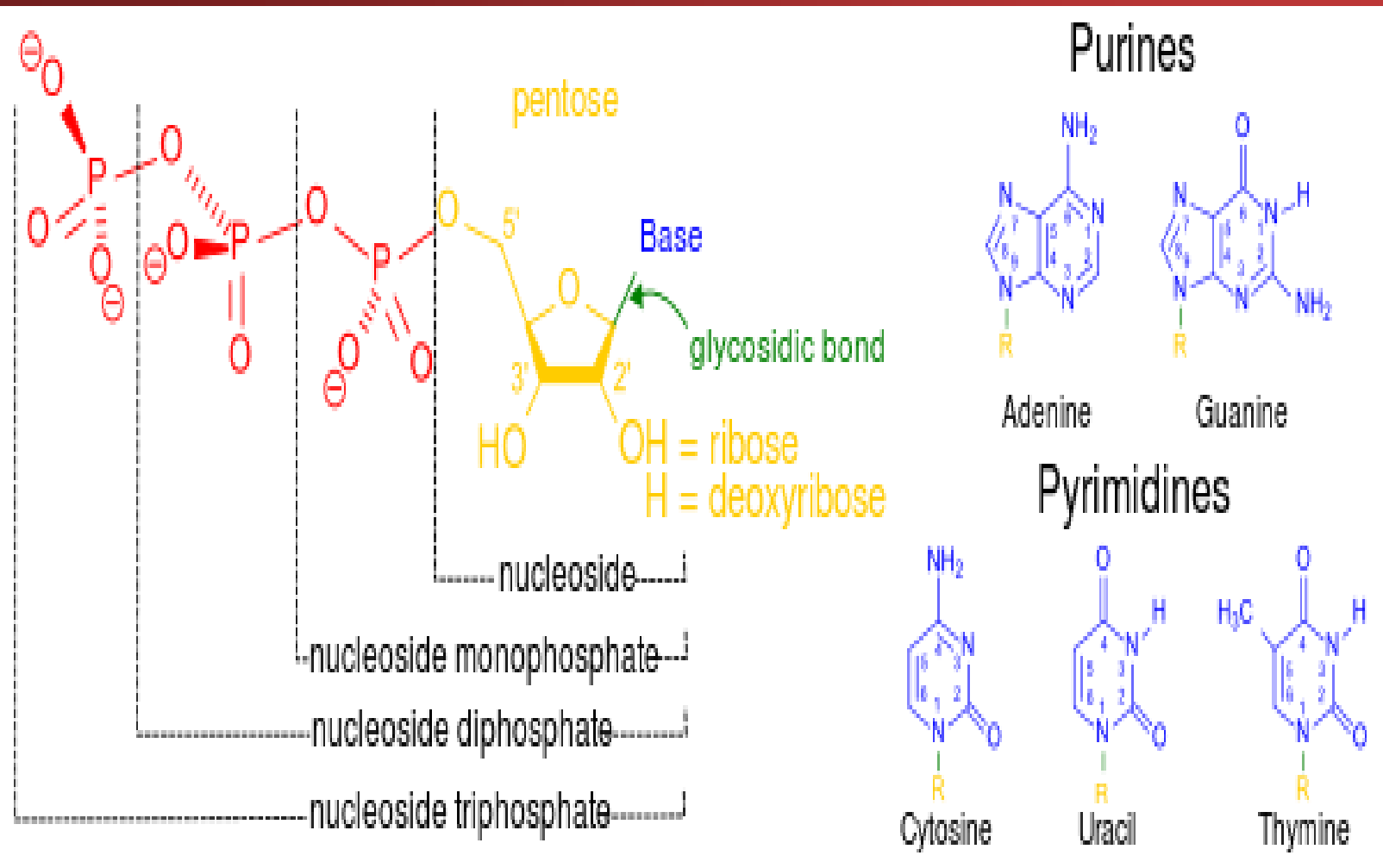
- Since the ribonucleosides have three free hydroxyl groups on their sugar ring, three possible monophosphate can be formed. Adenosine e.g. can give: Adenosine 5'-phosphate, Adenosine 3'-phosphate, Adenosine 2'-phosphate. In the same way guanosine, cytidine and uridine can give three monophosphate. The deoxyribonucleotides can only give two possible monophosphate.



4-Nucleotide structure

- In a nucleotide, a base is attached to a pentose sugar by N-glycosidic bonds to carbon # 1 of the sugar and a nitrogen atom of the base.
- Sugar is attached at position N-1 of the pyrimidine base.
- Sugar is attached at position N-9 of the purine base.
- The phosphate is attached to the 5' carbon of the sugar by phosphodiester linkages
- The ribo and deoxyribo nucleoside 5'-phosphate may be further phosphorylated at position 5' to yield 5'-di- and 5'-tri-phosphates. e.g. Adenosine 5'-phosphate (AMP), Adenosine 5'-diphosphate (ADP), Adenosine 5'-triphosphate (ATP). Mono, di- and tri-phosphates of ribonucleotides and deoxyribonucleotides all occur free state in the cell

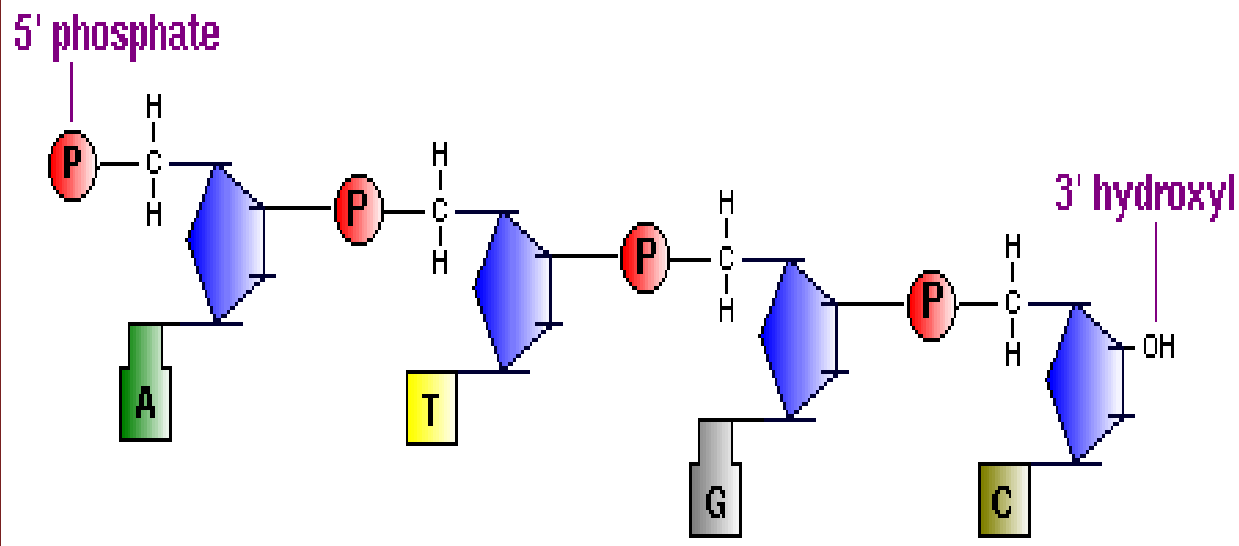
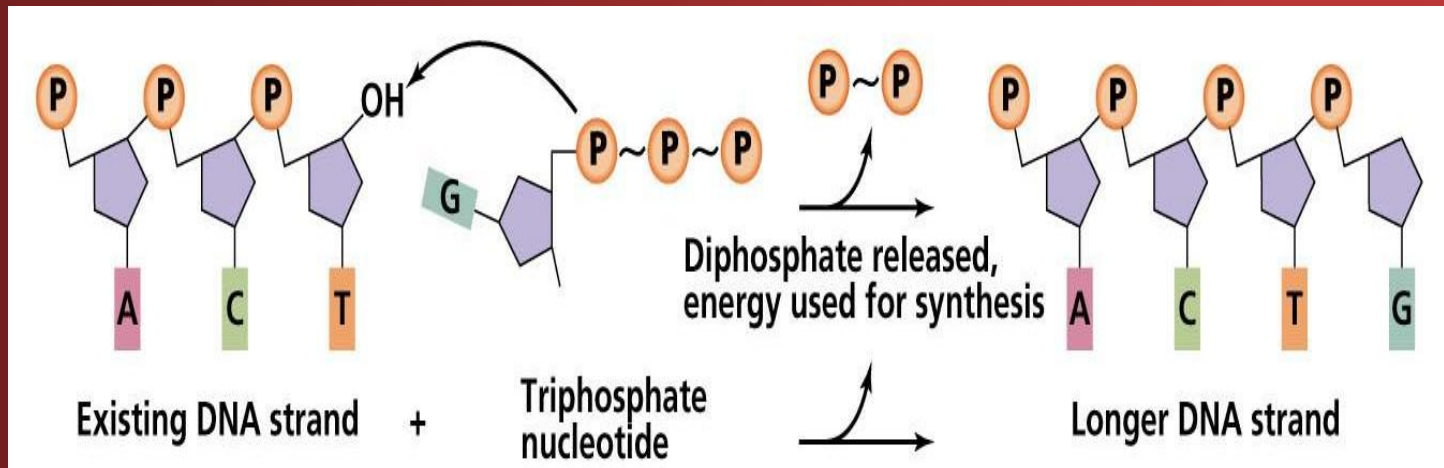
4-Nucleotide structure



5-Nucleotides polymerization

- Chemically, nucleic acids are composed of covalently linked chains of nucleotides that are attached by phosphate residues from carbon 5' of one sugar to OH group at carbon 3' of the adjacent sugar (phosphodiester bonds).
- Cohn and his colleagues (1956) provided full confirmation of the nature of the 5'-3'-internucleotide linkage in both DNA and RNA and made it possible for the clear concepts as to the **primary structure** of the two types of nucleic acids.
- The bases carry genetic information, the sugar and phosphates perform a structural role.

5-Nucleotides polymerization



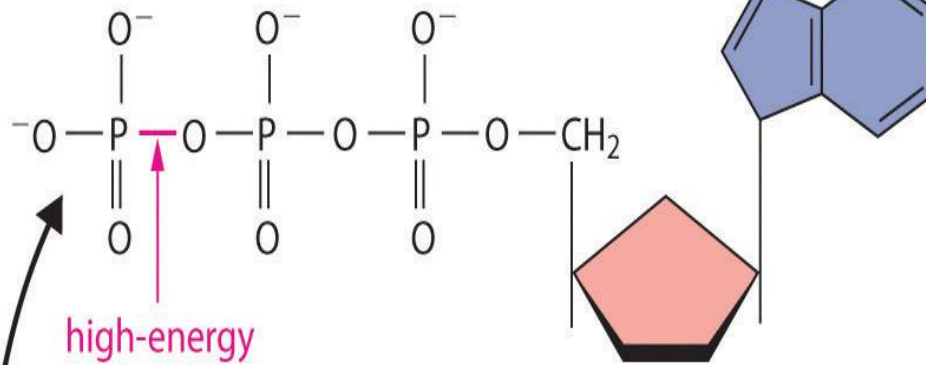
Functions of Nucleotides

- Cyclic nucleotides are **signal molecules** and regulators of cellular metabolism and reproduction
- **ATP** is central to energy metabolism
- **GTP** drives protein synthesis
- **CTP** drives lipid synthesis
- **UTP** drives carbohydrate metabolism

ATP

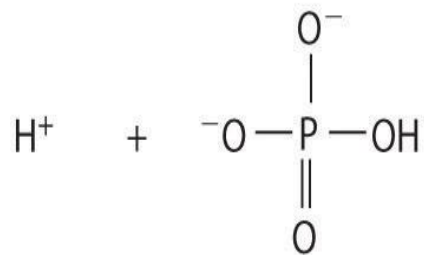
- ATP transports chemical energy within cells for metabolism.
- Most cellular functions need energy in order to be carried out:
 1. synthesis of proteins,
 2. synthesis of membranes,
 3. movement of the cell, cellular division,
 4. transport of various solutes etc.
- The ATP is the molecule that carries energy to the place where the energy is needed. When ATP breaks into ADP (Adenosine diphosphate) and P_i (phosphate), the breakdown of the last covalent link of phosphate (a simple - PO₄) liberates energy that is used in reactions where it is needed.

ATP

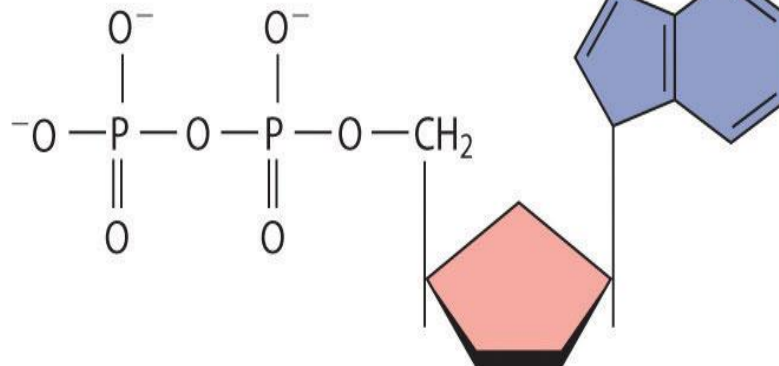


Energy from oxidation of nutrients is stored in high-energy bond.

$- \text{H}_2\text{O}$



phosphate



ADP

$+ \text{H}_2\text{O}$

Energy stored in high-energy bond is released.